EngageIQ — Smart Engagement Opportunity Scorer

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Live: https://engageiq-bax423-final.streamlit.app/ · Repo: https://github.com/shprasa/engageiq-bax423-final

1. Overview & local run

EngageIQ ranks GitHub engagement opportunities against interest profiles or four course personas. Streamlit dashboard: ranked cards, Why-this scores, Engage/Bookmark/Skip feedback, trend analytics, PDF/CSV export.

Grader local run (no API keys; offline data in ZIP). Unzip and copy/paste:

```
cd code
py -m pip install -r requirements.txt
py -m streamlit run app.py
```

Opens http://localhost:8501 (use python3 on macOS/Linux).

2. Data sources & offline dataset

Sources (≥2): GitHub REST API (scrape_github.py) + GitHub Archive (scrape_gharchive.py). Snapshot: 18,019 records (100% live API URLs), all 15 domains, in data/opportunities_snapshot.csv (6,177 GitHub API + 11,842 GitHub Archive). DuckDB loads on startup. Built by scripts/build_snapshot.py.

3. System architecture

Python 3.11 · Streamlit · scikit-learn · DuckDB. Flow: scrape → CSV → stream/dedup → DuckDB → embed → rerank → UI.

4. Six core capabilities (all required — missing any caps score at 60/100)

5. Pipeline design

6. BAX-423 technique choices & rationale

EngageIQ integrates two BAX-423 techniques into the core ranking loop: a recommendation pipeline for retrieval and multi-stage ranking, and reinforcement learning for adaptive personalization from feedback.

Technique 1 — Recommendation system: profiles and opportunity text are embedded with TF-IDF (128-d SVD), indexed with cosine NearestNeighbors for ANN retrieval (top-200), then reranked on relevance, community health, visibility, effort, and recency with persona-specific boosts. TF-IDF/SVD was chosen over deep embeddings because it runs fully offline in the ZIP, cold-starts for new profiles, and keeps Why-this scores interpretable. Ranking is benchmarked with NDCG@10 (simulated avg 1.00) and persona top-10 checks: Sofia 6/10 GFIs; David 10/10 DevOps; Raj 10/10 devtools; Lina 66% interest match.

Technique 2 — Reinforcement learning: Engage (+1.0), Bookmark (+0.85), and Skip (0.0) update a Thompson-sampling bandit over 15 domain arms; posterior samples shift domain weights in the reranker. This satisfies the 50+ feedback-round requirement without full deep RL. Over 60 simulated rounds, average reward in the last 10 improves from 0.23 to 1.00 (+0.78), showing the policy learns to deprioritize skipped domains (e.g., Raj low-engagement threads after simulated skips).

7. Test persona results (pass/fail table required)

Automated in persona_eval.py (exact PDF criteria). user_test_loop.py: 15/15 PASS.

Six capabilities × four personas (24/24 PASS):

8. Limitations

Two sources only (GitHub API + GH Archive); GH Archive threads proxy Reddit/blog discussion items in persona validation.

In-process streaming + DuckDB dedup, not Kafka/Spark. TF-IDF/SVD vs deep embeddings; hidden personas not pre-tested.

Suggested actions use offline templates unless optional LLM API keys configured.

9. Submission & deployment

ZIP: Prasad_Shivneel_BAX423_Final.zip (code/, data/, brief.pdf, prompts.md). Deploy: Streamlit Cloud → https://engageiq-bax423-final.streamlit.app/, entry code/app.py. Demo: Sofia GFIs → Engage/Skip RL → David DevOps → Lina WoW chart → Raj devtools → export brief.